7

第八周作业

习题 7.1: P223 Ex 5.5.3

设数域$\mathbb{F}$上n维线性空间V是子空间U与W的直和，则对任意$\alpha \in V$，存在唯一一对向量β与$\gamma, \beta \in U, \gamma \in W$，使得$\alpha = \beta + \gamma$. 定义映射$\mathscr{A} : V \to V$如下：设$\alpha \in V$，则令$\mathscr{A}(\alpha) = \beta$. 映射$\mathscr{A}$称为$V$沿子空间$W$在$U$上的投影变换。证明：

(1) 投影变换$\mathscr{A}$是线性变换；

(2) 线性变换$\mathscr{B} : V \to V$为投影变换的充分必要条件是$\mathscr{B}$为幂等变换，即$\mathscr{B}$满足$\mathscr{B}^2 = \mathscr{B}$；

(3) 线性变换$\mathscr{B} : V \to V$为投影变换的充分必要条件是$\mathscr{I} - \mathscr{B}$为投影变换，其中$\mathscr{I}$是单位映射。

习题 7.2: P223 Ex 5.5.8

设$\mathscr{A} : V \to V$与$\mathscr{B} : V \to V$是幂等变换，即$\mathscr{A}^2 = \mathscr{A}, \mathscr{B}^2 = \mathscr{B}$. 证明：

(1) $\mathrm{Im}(\mathscr{A}) = \mathrm{Im}(\mathscr{B})$的充分必要条件是$\mathscr{A}\mathscr{B} = \mathscr{B}, \mathscr{B}\mathscr{A} = \mathscr{A}$；

(2) $\mathrm{Ker}(\mathscr{A}) = \mathrm{Ker}(\mathscr{B})$的充分必要条件是$\mathscr{A}\mathscr{B} = \mathscr{A}, \mathscr{B}\mathscr{A} = \mathscr{B}$.

习题 7.3: P224 Ex 5.5.13

设$A \in \mathbb{C}^{n\times n}, f(x), g(x) \in \mathbb{C}[x]$，且$(f(\lambda), g(\lambda)) = 1$. 证明：

$$\mathrm{rank}\, f(A) + \mathrm{rank}\, g(A) = n + \mathrm{rank}\, f(A)g(A).$$

习题 7.4: P229 Ex 5.6.3

设$\mathscr{A} : V \to V$是线性变换，而$0 \neq \alpha \in V$，证明：V中向量$\alpha, \mathscr{A}(\alpha), \cdots, \mathscr{A}^k(\alpha), \cdots$生成的子空间$U$是$\mathscr{A}$的不变子空间。设$\dim U = r$，证明：$\{\alpha, \mathscr{A}(\alpha), \cdots, \mathscr{A}^{r-1}(\alpha)\}$是$U$的基。求$\mathscr{A}|_U$在这组基下的方阵。

习题 7.5: P229 Ex 5.6.5

设V是n维复线性空间，且$\mathscr{A}: V \to V$是线性变换。证明：V的每个子空间都是$\mathscr{A}$的不变子空间的充分必要条件是$\mathscr{A}$为纯量变换，即$\mathscr{A} = \lambda\mathscr{I}$，其中$\lambda \in \mathbb{C}$.